

Notes on MLS Chapter 4, Section 3.1 and 3.2

NJ Theron

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On p. 168 MLS write that the first i columns of $J_{sl_i}^b$ are calculated as

$$\xi_j^\dagger = \text{Ad}_{\left(e^{\hat{\xi}_j \theta_j} \dots e^{\hat{\xi}_i \theta_i} g_{sl_i}(0)\right)}^{-1} \xi_j \quad j \leq i. \quad (\text{a})$$

I am of opinion that this is incorrect, and incompatible with what MLS writes on p. 176. In my view, this should be replaced by

$$\xi_j^\dagger = \begin{cases} \text{Ad}_{\left(e^{\hat{\xi}_{j+1} \theta_{j+1}} \dots e^{\hat{\xi}_i \theta_i} g_{sl_i}(0)\right)}^{-1} \xi_j & j < i \\ \text{Ad}_{\left(g_{sl_i}(0)\right)}^{-1} \xi_j & j = i \end{cases} \quad (\text{b})$$

To support this argument, consider frames $L_j, L_{j+1} \dots L_i, j < i$.

$$g_{sl_j}(\theta) = e^{\hat{\xi}_1 \theta_1} \dots e^{\hat{\xi}_j \theta_j} g_{sl_j}(0)$$

$$g_{sl_{j+1}}(\theta) = e^{\hat{\xi}_1 \theta_1} \dots e^{\hat{\xi}_j \theta_j} e^{\hat{\xi}_{j+1} \theta_{j+1}} g_{sl_{j+1}}(0)$$

$$\begin{aligned} g_{sl_i}(\theta) &= e^{\hat{\xi}_1 \theta_1} \dots e^{\hat{\xi}_j \theta_j} e^{\hat{\xi}_{j+1} \theta_{j+1}} \dots e^{\hat{\xi}_i \theta_i} g_{sl_i}(0) \\ &= e^{\hat{\xi}_1 \theta_1} \dots e^{\hat{\xi}_j \theta_j} g_{ji} g_{sl_i}(0), \end{aligned}$$

where

$$g_{ji} = e^{\hat{\xi}_{j+1} \theta_{j+1}} \dots e^{\hat{\xi}_i \theta_i}$$

can be considered as the homogeneous transformation that generates frame L_i from frame L_j , expressed in frame L_j . From this it follows that

$$g_{ij} = e^{-\hat{\xi}_i \theta_i} \dots e^{-\hat{\xi}_{j+1} \theta_{j+1}},$$

of course still for the case where $j < i$. When $i = j$, $g_{ii} = I$. Therefore

$$g_{ij} = \begin{cases} e^{-\hat{\xi}_i \theta_i} \dots e^{-\hat{\xi}_{j+1} \theta_{j+1}} & i > j \\ I & i = j \end{cases},$$

which is in agreement with the first equation on MLS p. 177.

Now, the note “Composition of Adjoint Transformations.pdf”, available on Moolde under the “Notes” topic, shows that, for two consecutive homogeneous transformations $g_{sc} = g_{sb}g_{bc}$ (MLS eq. (2.24) p. 37),

$$\text{Ad}_{g_{sc}} = \text{Ad}_{g_{sb}} \text{Ad}_{g_{bc}} .$$

Using this, the first equation in equation (b) above, for the case where $j < i$, can be written as

$$\xi_j^\dagger = \text{Ad}_{(g_{sl_i}(0))}^{-1} \text{Ad}_{(e^{\hat{\xi}_{j+1}\theta_{j+1}} \dots e^{\hat{\xi}_i\theta_i})}^{-1} \xi_j = \text{Ad}_{(g_{sl_i}(0))}^{-1} \text{Ad}_{g_{ji}}^{-1} \xi_j = \text{Ad}_{(g_{sl_i}(0))}^{-1} \text{Ad}_{g_{ij}} \xi_j = \text{Ad}_{(g_{sl_i}(0))}^{-1} A_{ij} \xi_j ,$$

where A_{ij} is defined in eq. (4.27) p. 176. Note, that

$$\text{Ad}_{g_{ij}} = A_{ij}$$

is pointed out by MLS at the top of p. 177, just below the g_{ij} equation.

In my opinion, therefore, it is clear that $e^{\hat{\xi}_j\theta_j}$ can indeed be used to calculate $g_{sl_j}(\theta)$, but it does not play a role in defining the j th instantaneous joint twist relative to the i th link frame, and that eq. (a) on the previous page is therefore incorrect (and so also the corresponding equation on MLS p. 168).

Also, on MLS p. 176, there is, in my view, an error at the end of the second line of text following eq. (4.27): the $\text{Ad}_{g_{sl_i}}^{-1} A_{ij} \xi_j$ should be replaced with $\text{Ad}_{g_{sl_i}(0)}^{-1} A_{ij} \xi_j$.